Chronic child malnutrition trends in Peru

"Chronic child malnutrition (stunting) trends and regional inequalities in Peru after 2015, including stagnation of national prevalence and rural-urban or department-level gaps. Return a matrix with: authors, year, journal, DOI, data source (ENDES or other), period, prevalence reported, and main determinants. Limit to 2016-2026." *Generated by LeapSpace*

Peru's **national stunting prevalence kept falling after 2015 but then appeared to plateau around 12% from 2018 onward**, while **large rural-urban and department-level gaps persisted**—with rural prevalence consistently far above urban, and high-burden departments such as Huancavelica, Loreto, and several Andean/Amazonian areas remaining well above the national average [1] [2] [3] [4] [5].

Matrix: Peru stunting trends and inequalities, 2016–2026

Authors	Year	Journal	DOI	Data source	Period	Prevalence reported	Main determinants
Sobrino et al. / ***Chronic Malnutrition among Children under Five in Peru: A Spatial Analysis of Nutritional Data, 2010-2016***	2017	Revista española de salud pública	Not stated in retrieved record	Nutritional Status Information System (SIEN)	2010 vs 2016	**23.9%** in 2010 and **18.0%** in 2016; rural prevalence fell by **7.6 percentage points**; only Huancavelica remained above 30% [4]	Spatial clustering concentrated in the **sierra and jungle**; persistent **regional heterogeneity** rather than uniform decline [4]
Huicho et al. / ***Drivers of stunting reduction in Peru: A country case study***	2020	American Journal of Clinical Nutrition	10.1093/ajcn/nqaa164	DHS / Peru survey data; department-level multilevel analysis	2000–2016	National stunting fell from **31.3% to 13.1%** [6]	Higher **maternal BMI**, **maternal height**, improved **maternal/newborn care**, **parental education**, **urban migration**, and **reduced fertility** were the main contributors; reduction was stronger among the **poorest**, **rural**, and least educated groups [6]
Not stated in retrieved snippet / ***Relationship between stunting in children 6 to 36 months of age and maternal employment status in Peru***	2019	PLoS ONE	10.1371/journal.pone.0212164	Peruvian DHS / ENDES	Survey year not explicitly stated in snippet	Stunting prevalence **15.9%** in children 6–36 months [7]	**Unpaid maternal work** was associated with higher stunting risk; overall maternal employment status was not associated [7]
Not stated in retrieved snippet / ***The double burden of malnutrition in Peru: An update with a***	2021	American Journal of Clinical Nutrition	10.1093/ajcn/nqaa387	Peruvian nationally representative surveys / ENDES-DHS	1992–2017	Stunting decreased overall; **urban AAPC -4.5%** vs **rural AAPC -2.6%**; inequities by in-	**Socioeconomic status** and **urban-rural residence** shaped trends; disparities widened even as

focus on social inequities						come and residence widened [8]	national prevalence fell [8]
Not stated in retrieved snippet / ***Stunting in preschool and school-age children in the Peruvian highlands and its association with Fasciola infection and demographic factors***	2021	PLoS Neglected Tropical Diseases	10.1371/journal.pntd.0009519	Study population in the Peruvian highlands; comparison with ENDES 2017 and rural/urban figures cited	Study population; comparisons to 2017 ENDES	**21.3%** in highlands; some districts reached **32%**; rural **25.7%** vs urban **7.3%** [9]	**Poverty score**, fewer **years of parental education**, and **higher home elevation** were associated with lower HAZ; the paper also highlights **rural disadvantage** [9]
Not stated in retrieved snippet / ***Malnutrition in all its forms and socio-economic disparities in children under 5 years of age and women of reproductive age in Peru***	2020	Public Health Nutrition	10.1017/S136898001900315X	National Demographic and Family Health Survey	2015	Stunting was **significantly lower** among children with **high SES** and **high-educated mothers**, and **higher** among **indigenous children** [10]	**Wealth**, **maternal education**, and **ethnicity** explained major disparities; inequities in malnutrition were clearly patterned by social position [10]
Not stated in retrieved snippet / ***Urbanization in Peru is inversely associated with double burden of malnutrition***	2021	Obesity	10.1002/oby.23188	Pooled DHS / ENDES mother-child pairs	2009–2019	Stunting fell from **23.8% to 12.2%** nationally; rural decline was slower than urban [11]	**Urbanization** was inversely associated with the child undernutrition burden; the paper emphasizes the slower rural decline [11]
Not stated in retrieved snippet / ***Trends in stunting according to regions of Peru and sex: Period 2007-2018***	2023	Revista del Cuerpo Medico Hospital Nacional Almanzor Aguinaga Asenjo	Not stated in retrieved record	ENDES	2007–2018	National stunting fell by **17 percentage points**; reduction was significant in **22 of 24 regions** [3]	Uneven regional performance: **Tacna, Cusco, Madre de Dios** did best; **La Libertad, Cajamarca, Loreto** lagged; stunting was more frequent in **boys** and in **rural areas**, especially in the **mountains** [3]
Not stated in retrieved snippet / ***Exploring the magnitude and drivers of the double burden of malnutrition at maternal and dyad levels in peri-urban Peru***	2023	Maternal and Child Nutrition	10.1111/mcn.13549	Peruvian surveys / INEI estimates cited	1992–2021	Stunting decreased from **36.6% in 1992 to 11.5% in 2020**; urban declined faster than rural [12]	The paper attributes the faster urban decline to **urbanicity-related differences** and notes persistent inequality by residence [12]
Not stated in retrieved snippet / ***A spatial analysis of disparity in the prevalence of stunting rates among children aged under five between rural and urban areas in Peru***	2025	Letters in Spatial and Resource Sciences	10.1007/s12076-025-00407-0	ENDES 2013, 2018, 2023	2013–2023	Despite reduction in all regions, **rural-urban disparities remained large** [5]	The study focuses on **small-area spatial disparity** and the limitation of rural underrepresentation in direct survey estimates [5]
Not stated in retrieved snippet / ***Modelo Predictivo de Riesgo de Desnutrición Crónica Infantil en el Perú medi-***	n.d.	Uploaded document	Not stated in retrieved record	ENDES 2007–2025	2007–2025	National prevalence **29.2% in 2007**, **13.0% in 2017**, then **11.3%–12.4% from 2018 onward**, reaching **12.0% in 2025**;	Strongest predictors were **household poverty**, **birth weight**, **altitude**, and **birth size**; territory-specific patterns dif-

ante Machine Learning						rural **21.5%** vs urban **7.9%** in 2025 [1]	fered by department (e.g., Puno and Huancavelica) [13]

Bottom line

The post-2015 Peruvian pattern is **declining but increasingly unequal**: national stunting dropped into the low teens, then **stagnated around 12% after 2018**, while **rural children remained about two to three times more affected than urban children** and several **Andean and Amazonian departments** continued to concentrate the highest burdens [1] [2] [3] [5]. Across studies, the most consistent determinants are **poverty, low maternal education, rural residence, altitude/territorial disadvantage, and weaker access to health and sanitation-related conditions** [6] [8] [10] [13].

Confidence level: **High** — multiple Peru-specific studies using ENDES/DHS, SIEN, and related national analyses consistently show the same overall direction of national decline with persistent rural-urban and regional inequalities [1] [3] [4] [6] [8].

References

1. Uploaded file: main.pdf
2. Uploaded file: main.pdf
3. Aparco-Balboa, J.P., Morales-Cahuancama, B., Alvis-Chirinos, K., Solis-Sánchez, G. (2023) Trends in stunting according to regions of Peru and sex: Period 2007-2018. *Revista del Cuerpo Medico Hospital Nacional Almanzor Aguinaga Asenjo*. https://www.scopus.com/pages/publications/85173923280
4. Hernández-Vásquez, A., Tapia-López, E. (2017) Chronic Malnutrition among Children under Five in Peru: A Spatial Analysis of Nutritional Data, 2010-2016. *Revista espanola de salud publica*. https://www.sciencedirect.com/science/article/pii/e201705035
5. Sikov, A., Cerda-Hernández, J. (2025) A spatial analysis of disparity in the prevalence of stunting rates among children aged under five between rural and urban areas in Peru. *Letters in Spatial and Resource Sciences*. https://doi.org/10.1007/s12076-025-00407-0
6. Huicho, L., Vidal-Cardenas, E., Akseer, N., (...), Bhutta, Z.A. (2020) Drivers of stunting reduction in Peru: A country case study. *American Journal of Clinical Nutrition*. https://doi.org/10.1093/ajcn/nqaa164

7. Chávez-Zárate, A., Maguiña, J.L., Quichiz-Lara, A.D., (...), Mayta-Tristán, P. (2019) Relationship between stunting in children 6 to 36 months of age and maternal employment status in Peru: A sub-analysis of the Peruvian Demographic and Health Survey. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0212164
8. Santos, M.P., Turner, B., Chaparro, M.P. (2021) The double burden of malnutrition in Peru: An update with a focus on social inequities. *American Journal of Clinical Nutrition*. https://doi.org/10.1093/ajcn/nqaa387
9. Webb, Camille M., Morales, Maria Luisa, Lopez, Martha, (...), Cabada, Miguel M. (2021) *PLoS Neglected Tropical Diseases*. https://doi.org/10.1371/journal.pntd.0009519
10. Curi-Quinto, K., Ortiz-Panozo, E., de Romaña, D.L. (2020) Malnutrition in all its forms and socio-economic disparities in children under 5 years of age and women of reproductive age in Peru. *Public Health Nutrition*. https://doi.org/10.1017/S136898001900315X
11. https://www.sciencedirect.com/science/article/pii/S1930739X21000032
12. https://www.sciencedirect.com/science/article/pii/S1740870923000771
13. Uploaded file: main.pdf